TechCo SQL and Database Standards

Naming

Table names are plural and written in snake_case, for example customer_orders.

Every table has a primary key column named id, which must be unique and never null.

Foreign key columns end in _id, such as customer_id or order_id.

Writing Queries

Use the HAVING clause to filter groups after GROUP BY, and WHERE to filter rows before grouping.

Use an INNER JOIN when you need only rows that match in both tables.

Use DISTINCT only when duplicates are expected; it hides join mistakes otherwise.

Never use SELECT * in application code; list the columns you need.

Queries that run on every page load must finish in under 200 milliseconds.

Indexes and Performance

Add an index to any column used in a WHERE clause on a table with more than 10000 rows.

Check slow queries with EXPLAIN before asking for a review.

Large reports run on the read replica, never on the primary database.

Migrations

Every schema change is a migration file reviewed in a pull request.

Migrations must be reversible, so each one includes a down step.

Do not rename a column in one step; add the new column, copy the data, then drop the old column.